

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	1	1	3	4	3	2	1	4	4	4	2	1	1	3	1	4	3	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	2	1	1	2	4	2	2	1	2	4	3	3	1	4	2	4	1	2	3
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	3	2	1	3	2	1	1	4	2	3	1	4	2	3	1	4	1	1	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	2	4	3	4	4	4	2	4	1	2	4	3	4	4	4	4	4	2	3	4
Q.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	2	2	4	2	1	4	1	2	3	2	1	3	4	3	2	4	3	2	4	4
Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	3	2	4	4	4	1	3	4	3	4	1	4	3	3	3	4	3	2	2
Q.	121	122	123	124	125	126	127	128	129	130										
A.	3	3	4	3	2	3	4	2	2	3										

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HINT – SHEET

1. Ans (2)

$$V_{CM} = \frac{v}{2}$$

$$V'_A = \frac{3}{2} \times \frac{V}{2} - V = -\frac{V}{4}$$

$$V'_B = \frac{3V}{4}$$

$$\vec{\tau} = \vec{P}_f - \vec{P}_i = \frac{3mv}{4}$$

$$\Delta KE = \frac{1}{2} \left(\frac{m}{2} \right) v^2 \left(1 - \frac{1}{4} \right)$$

$$= \frac{3mv^2}{16}$$

2. Ans (3)

$$v = 2t$$

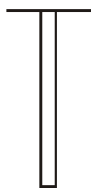
$$a_t = 2 \text{ m/s}^2$$

$$a_c = \frac{v^2}{r} = \frac{4t^2}{20} = \frac{t^2}{5} = \frac{25 \times 3}{50} = \frac{3}{2} \text{ m/s}^2$$

$$\text{So } a_{\text{total}} = \sqrt{4 + \frac{9}{4}} = \sqrt{\frac{25}{4}} = 2.5 \text{ m/s}^2$$

3. **Ans (1)**

(I) Tension and stress are maximum at point of suspension



$$T = mg = \rho A \ell g$$

$$\frac{T}{A} = \rho \ell g = \sigma_B$$

$$\ell = \frac{\sigma_B}{\rho g}$$

4. **Ans (1)**

$$\frac{dU}{dA} = T$$

$dU \rightarrow$ Surface energy or work done

$dA \rightarrow$ Change in Area

$$\begin{aligned} dU = \text{work done} &= [4\pi[2r]^2 - 4\pi r^2] T \times 2 \\ &= 2 \times 12\pi r^2 \times T \\ &= 24\pi r^2 T \end{aligned}$$

5. **Ans (3)**

$$\frac{dA}{dt} = \frac{L}{2m} = \text{constant}$$

$$\text{So, } \frac{A_{AB}}{t_{AB}} = \frac{A_{CD}}{t_{CD}}$$

$$\Rightarrow \frac{A}{T_1} = \frac{2A}{T_2}$$

$$T_1 = 0.5 T_2$$

6. **Ans (4)**

$$\Delta T = T, \alpha = \frac{\gamma}{3}$$

$$\text{Strain} = \frac{\Delta \ell}{\ell} = \alpha \Delta T$$

$$E = \frac{1}{2} \times \text{stress} \times \text{strain}$$

$$= \frac{1}{2} Y \alpha^2 \Delta T^2$$

7. **Ans (3)**

$$PV = nRT \quad \dots(i)$$

$$P \Delta V = nR \Delta T \quad \dots(ii) \text{ (P = constant)}$$

(ii)/(i)

$$\frac{\Delta V}{V} = \frac{\Delta T}{T} \Rightarrow \frac{\Delta V}{V \Delta T} = \frac{1}{T}$$

$$\text{So } \delta = \frac{1}{T} \left(\frac{\Delta V}{V \Delta T} = \delta \right)$$

So graph will be rectangular hyperbola.

8. **Ans (2)**

$$\Delta T \propto \frac{1}{K}$$

K same $\Rightarrow \Delta T$ same

9. **Ans (1)**

Wein's Displacement Law.

10. **Ans (4)**

ΔU does not depends on path, work done by gas depends on path.

11. **Ans (4)**

Work done by gas, is area enclosed by cycle.

$$w = \pi ab$$

$$= \pi 4 \times 4 = 16\pi \text{ lt - atm}$$

12. **Ans (4)**

$$\frac{n_2 R(300)}{v} = \frac{n_1 R(150)}{8v}$$

$$m_2 = \frac{m}{16}$$

13. **Ans (2)**

$$\text{As } TV^{\gamma-1} = \text{constant} \Rightarrow T \propto V^{1-\gamma}$$

$$\Rightarrow \frac{dT}{T} = (1-\gamma) \frac{dV}{V} \Rightarrow \frac{dV}{dT} = \frac{-V}{(\gamma-1)T} = \tan 45^\circ = 1$$

$$\Rightarrow \gamma = 1 + \frac{V_0}{T_0}$$

Bulk modulus

$$K = \frac{dP}{-dV/V} = \gamma P = \left(1 + \frac{V_0}{T_0}\right) P_0 = \left(1 + \frac{V_0}{T_0}\right) \left(\frac{2RT_0}{V_0}\right)$$

$$= 2R \left(1 + \frac{T_0}{V_0}\right)$$

14. **Ans (1)**

Here $T_1 = 400 \text{ K}$, $Q_1 = 100 \text{ kcal}$, $Q_2 = 70 \text{ kcal}$

$$(i) \frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$

$$\therefore T_2 = \frac{Q_2}{Q_1} T_1 = \frac{70}{100} \times 400 = 280 \text{ K}$$

$$(ii) \eta = 1 - \frac{T_2}{T_1} = 1 - \frac{280}{400} = 1 - 0.7 = 0.3$$

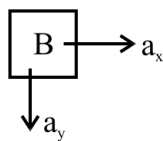
$$\text{or } \% \eta = 0.3 \times 100 = 30\%$$

15. **Ans (1)**

Use standard equation of trajectory

16. **Ans (3)**

Vertical component of acceleration of B will be equal to acceleration of A



17. **Ans (1)**

$$I_1 = \frac{m\ell^2}{3} + m\ell^2 = \frac{4}{3}m\ell^2$$

$$I_2 = \frac{m\ell^2}{3}$$

$$I_3 = \frac{m\ell^2}{12}$$

$$I_4 = \frac{m\ell^2}{3} + m\left(\frac{\ell}{2}\right)^2 = \frac{m\ell^2}{3} + \frac{m\ell^2}{4}$$

$$I_4 = \frac{7m\ell^2}{12}$$

$$I_1 > I_4 > I_2 > I_3$$

18. **Ans (4)**

$$[F] = \left[\frac{Cr}{V^2} \right]$$

$$[M^1 L^1 T^{-2}] = \left[\frac{CL}{L^2} T^2 \right]$$

$$[C] = [M^1 L^2 T^{-4}]$$

19. **Ans (3)**

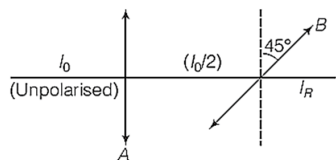
$$\frac{\lambda}{2} = 1 \Rightarrow \lambda = 2 \Rightarrow v = \frac{\omega}{k} \Rightarrow \frac{\lambda}{T} = \frac{2}{2} \Rightarrow 1 \text{ cm/sec}$$

$$v_p = \omega \sqrt{A^2 - x^2} = \frac{2\pi}{T} \sqrt{4^2 - (2\sqrt{3})^2} = \frac{2\pi}{2} \sqrt{16 - 12}$$

$$\frac{2\pi}{2} \times 2 = 2\pi \text{ cm/sec}$$

20. **Ans (3)**

Relation between intensities is



$$I_R = \left(\frac{I_0}{2} \right) \cos^2(45^\circ) = \frac{I_0}{2} \times \frac{1}{2} = \frac{I_0}{4}$$

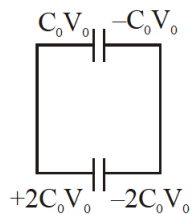
21. **Ans (1)**

The cell will deliver maximum power to the network if $R_{eq} = r$

$$\Rightarrow \frac{3R}{2} = r$$

$$\therefore R = \frac{2r}{3} = \frac{2}{3} \Omega$$

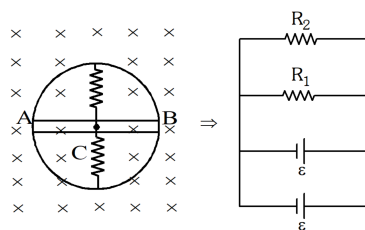
22. Ans (2)



$$Q = 3C_0 V_0 = (KC_0 + 2C_0)V$$

$$V = \frac{3}{K+2} V_0$$

23. Ans (1)



$$I_1 = \frac{\varepsilon}{R_1} = \frac{B\omega r^2}{2R_1}$$

24. Ans (1)

25. Ans (2)

$$U = -PE_{11}$$

$$= -P_1 \times \frac{2KP_2 \cos \theta}{r^3}$$

$$U = \frac{-2Kp_1 p_2 \cos \theta}{r^3}$$

26. Ans (4)

Conceptual Question

27. Ans (2)

$$\phi > \phi_c$$

$$\sin \phi > \sin \phi_c$$

$$\cos \alpha > \frac{1}{n} \quad \dots (i)$$

$$\sin \theta = n \sin \alpha$$

$$\frac{\sin \theta}{n} = \sin \alpha \quad \dots (ii)$$

$$\sin \theta < \sqrt{n^2 - 1}$$

$$\theta < \sin^{-1} \sqrt{n^2 - 1}$$

28. Ans (2)

The number of photons emitted per second is given by

$$= \frac{\text{Power of the source}}{\text{energy of each photon}} = \frac{P}{h\nu} = \frac{P\lambda}{hc}$$

These photons spread out in all directions over an area $4\pi r^2$. So, that the share of an area A is a fraction $\frac{A}{4\pi r^2}$ of the total number of photons emitted per second.

Hence, the number of photons striking the surface per second is given as

$$\frac{P\lambda}{hc} \cdot \frac{A}{4\pi r^2}$$

Thus, the option (3) is correct.

Further, the maximum energy of the emitted photoelectron is

$$(E_k)_{\max} = hc - h\nu_0 = \frac{hc}{\lambda} - \phi$$

Thus, the option (1) is also correct.

The stopping potential V_s is obtained from the equation

$$(E_k)_{\max} = \frac{1}{2}mv_{\max}^2 = eV_s$$

$$V_s = \frac{(E_k)_{\max}}{e} = \frac{1}{e} \left[\frac{hc}{\lambda} - \phi \right]$$

Thus, the option (2) is incorrect.

Also, photoemission occurs only for the wavelengths below the threshold wavelength λ_0 . Hence, the option (4) is also correct.

30. Ans (2)

When a charged particle is moving at right angles to the magnetic field then a force acts on it which behaves as a centripetal force and moves the particle in circular motion.

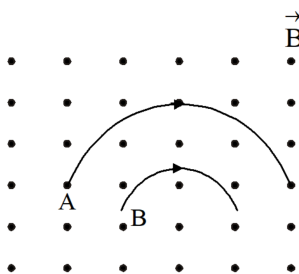
$$\therefore \frac{m_A v_A^2}{r_A} = q \cdot v_A B$$

$$\therefore \frac{m_A v_A}{r_A} = qB$$

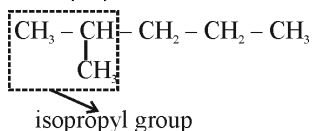
$$\frac{m_B v_B}{r_B} = qB$$

$$\Rightarrow \frac{m_A v_A}{r_A} = \frac{m_B v_B}{r_B}$$

$$\text{Since } r_A > r_B \Rightarrow m_A v_A > m_B v_B$$

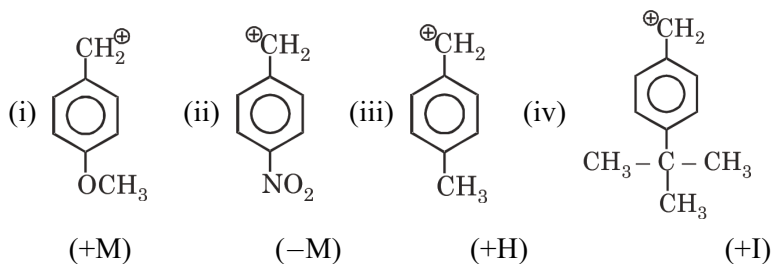


31. **Ans (4)**



32. **Ans (3)**

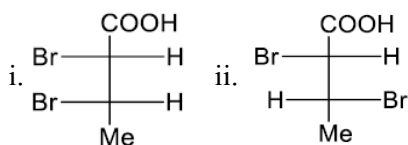
Correct order for stability of intermediate is :



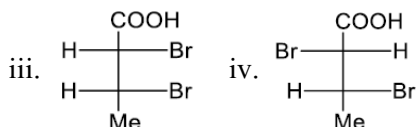
Order : (i) > (iii) > (iv) > (ii)

33. **Ans (3)**

The Fischer projection of (i) and (ii) is wrong because the functional group is not at the top. On rotating (i) and (ii) by 180°, they are represented as shown:



Now compare i, ii, iii and iv.



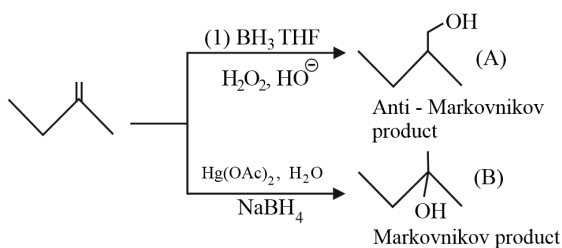
(i) and (iii) $\Rightarrow$ Enantiomers

(ii) and (iv) $\Rightarrow$ Identical or same

Therefore, the pairs of diastereomers are:

(i), (ii); (i), (iv); and (ii), (iii)

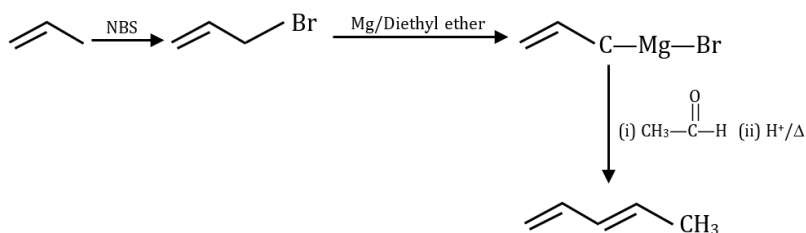
34. **Ans (1)**



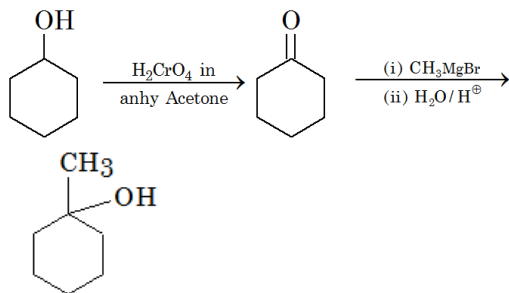
35. **Ans (4)**

Due to steric hindrance and Bredt's rule, compound D undergoes neither S_N2 nor S_N1 reaction.

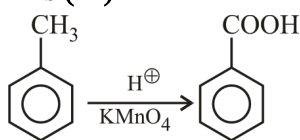
36. Ans (2)



37. Ans (4)

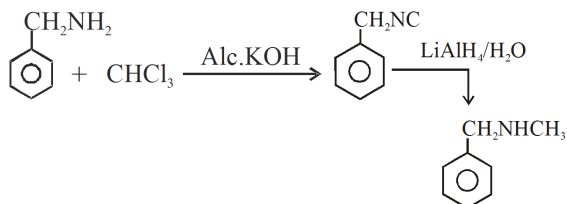


38. Ans (1)



$\therefore$ α -hydrogen is present side chain oxidised into $-\text{COOH}$ group.

39. Ans (2)



40. Ans (3)

(Nitrating mixture adds NO_2 at activated and less hindered position)

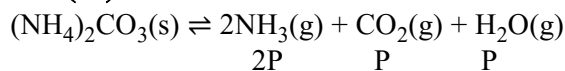
41. Ans (4)

$$\begin{array}{c}
 \text{O}_3 \quad \text{O}_2 \\
 \text{Mole ratio : } 1 \quad 2 \\
 [\text{VD}]_{\text{mix}} = \frac{M_{\text{avg}}}{2} = \frac{48 + 64}{2 \times 3} = \frac{112}{2 \times 3} = \frac{56}{3}
 \end{array}$$

42. Ans (3)

$$\begin{array}{l}
 E = -\frac{(\text{IP})}{n^2} \\
 \Delta E_{(1-3)} = \text{IP} \left(1 - \frac{1}{9} \right) = \text{IP} \times \frac{8}{9} = 36 \times \frac{8}{9} \\
 \Delta E = 32 \text{ eV}
 \end{array}$$

43. **Ans (2)**



$$K_p = (2P)^2 \cdot P \cdot P = 64 \text{ atm}$$

$$P = 2 \text{ atm}$$

$$P_{\text{total}} = 2P + P + P = 8 \text{ atm}$$

44. **Ans (1)**

From given data $t_{1/2} = 30 \text{ minute}$

$$K = \frac{\ln 2}{30} \quad r_t = 60 = k[A]_{t=60}$$

$$= \frac{\ln 2}{30} \times \frac{0.05}{2} = \frac{\ln 2}{1200}$$

45. **Ans (3)**

$$X \times 0.6 \times \frac{60}{100} = 200 \times 0.9 \times \frac{12}{100}$$

$$X = 60 \text{ mL}$$

46. **Ans (2)**

$$\Lambda_{\text{eq}ACo_4} = \Lambda_{ACoNa} + \Lambda_{HCl} - \Lambda_{NaCl}$$

47. **Ans (1)**

B, D : ideal solution

C : Negative deviation

48. **Ans (1)**

Correct option is A)

Y occupies CCP lattice hence the effective number of Y = 4 (effective no of atoms in CCP is 4)

X occupies $2/3$ of tetrahedral voids . Number of tetrahedral voids generated in any unit cell is = $2Z$ (twice of an effective number of atoms in unit cell)

$$= 2 \times 4 = 8$$

Since X occupies $2/3$ of tetrahedral voids , so Effective number of X = $\frac{2}{3} \times 8 =$

$$\frac{16}{3}$$

$$\text{So } X = \frac{16}{3} \text{ \& } Y = 4,$$

Hence simplest formula is X_4Y_3

49. **Ans (4)**

$$\frac{x}{m} = kP^{\frac{1}{n}} \text{ (Freundlich Adsorption isotherm)}$$

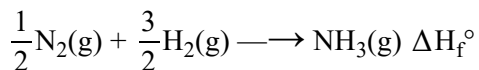
$$\frac{x}{m} = ?$$

$$\text{Log } \frac{x}{m} = \log k + \frac{1}{n} \log P \quad k = 10$$

$$\text{Slope} = \frac{1}{n} = \frac{1}{2}$$

$$\boxed{\frac{x}{m} = 7}$$

50. **Ans (2)**



$$\therefore \Delta H_f^\circ = \Delta H_{\text{Reaction}}^\circ \Sigma (\text{Bond enthalpies of Reactants}) - \Sigma (\text{B.E. of Products})$$

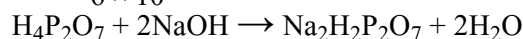
$$\Delta H_f^\circ = \left[\frac{1}{2}(\epsilon_{\text{N} \equiv \text{N}}) + \frac{3}{2}(\epsilon_{\text{H}-\text{H}}) \right] - (3 \times \epsilon_{\text{N}-\text{H}})$$

$$\Delta H_f^\circ = \frac{1}{2} \times x_1 + \frac{3}{2}x_2 - 3x_3$$

51. **Ans (3)**

$$n_{\text{H}_4\text{P}_2\text{O}_7} = \frac{534}{178} = 3$$

$$n_{\text{NaOH}} = \frac{30 \times 10^{23}}{6 \times 10^{23}} = 5$$



$$\frac{3}{1} \quad \frac{5}{2}$$

$$2.5 \quad 5$$

$$\therefore \text{NaOH is L.R}$$

$$\text{total mole of products} = 2.5 + 5$$

$$= 7.5$$

$$\text{total number of molecule present} = 7.5N_A$$

52. **Ans (1)**

$$\text{Given } \gamma = 4/3 \text{ or } 1.33$$

For reversible adiabatic expansion involving ideal gas

$$T_1 \cdot V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$300 \cdot V^{1/3} = T_2 \cdot (8V)^{1/3}$$

$$T_2 = 150 \text{ Kelvin}$$

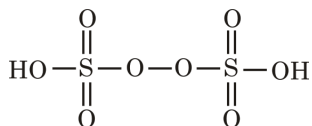
$$W = \Delta u = nC_V \cdot \Delta T$$

$$= 1 \times 3R \times (150 - 300)$$

$$= -450 R$$

$$|W| = 900 \text{ cal}$$

53. **Ans (4)**



54. **Ans (2)**

EN

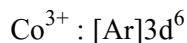
$$\text{Si} \quad 1.8$$

$$\text{P} \quad 2.1$$

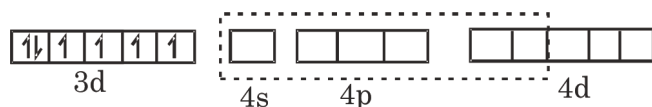
$$\text{C} \quad 2.5$$

$$\text{N} \quad 3.0$$

55. **Ans (3)**



This complex is outer orbital complex

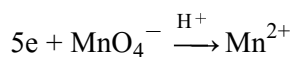


$$n = 4 \quad \mu = \sqrt{4(4+2)} \approx 4.9 \text{ B.M.}$$

Geometry : octahedral

56. **Ans (1)**

KMnO_4 is an oxidizing agent



$\text{Fe}_2(\text{SO}_4)_3$ cannot loose e^-]

57. **Ans (4)**

Photochemical smog occurs in warm, dry and sunny climate. The main components come from the action of sunlight on unsaturated hydrocarbon and nitrogen oxides produced by automobiles and factories.

58. **Ans (1)**

In Calgon process water soluble complex is formed with $\text{Na}_6\text{P}_6\text{O}_{18}$.

59. **Ans (1)**

Galena $\rightarrow$ PbS

Tinstone $\rightarrow$ SnO_2

Calamine $\rightarrow$ ZnCO_3

Bauxite $\rightarrow$ $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$

60. **Ans (3)**



71. **Ans (4)**

Tuberculosis is a disease of lungs. Similarly, Cataract is a disease of eyes.

72. **Ans (3)**

Squares of successive digits of number on the left hand side = Number on the right hand side

$$(4+1)^2 = 5^2 = 25$$

$$(6+1)^2 = 7^2 = 49$$

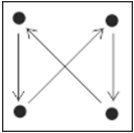
$$\text{Similarly } (2+1)^2 = 3^2 = 9$$

$$(3+1)^2 = 4^2 = 16$$

Hence 916 will be the value of question mark .

So, option (3) is correct option.

73. **Ans (4)**



The elements move in the sequence. The elements reaching the lower-left and the lower-right positions get rotated through 90° ACW while the elements reaching the upper-left and the upper-right positions get rotated through 90° CW.

74. **Ans (4)**

The logic followed here is:

$$4 \rightarrow 4^2 = 16 \rightarrow 4^3 = 64$$

$$6 \rightarrow 6^2 = 36 \rightarrow 6^3 = 216$$

$$8 \rightarrow 8^2 = 64 \rightarrow 8^3 = 512$$

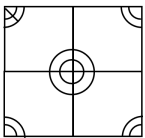
$$\text{Similarly, } 10 \rightarrow 10^2 = 100 \rightarrow 10^3 = 1000$$

Hence, 100, 1000 is the correct answer.

75. **Ans (4)**

In the other figures, the sequence of letters is in alphabetical order starting at the top and going clockwise.

76. **Ans (4)**



77. **Ans (4)**



Here the hairs are getting coloured when moving forward, and the next two figures to the figure of which of the hairs are getting coloured are changing their pattern of smile in clock-wise direction.

78. **Ans (2)**

The given series follows a logic that

$$11 \times 12, 12 \times 13, 13 \times 14, 14 \times 15, 15 \times 16, \dots$$

So, the missing number is $13 \times 14 = 182$.

79. **Ans (3)**

Take a look at the below series :

$$1500 - 25 = 1475$$

$$1475 - 27 = 1448$$

$$1448 - 29 = 1419$$

$$1419 - 31 = 1388$$

$$1388 - 33 = 1355$$

$$1355 - 35 = 1320$$

It clearly shows 1389 is wrong number.

80. **Ans (4)**



81. **Ans (2)**

In the second series, the image becomes simpler as the series proceeds. The trapezium inside the cloud loses its side one by one as the sequence proceeds one step forward to the right side.

82. **Ans (2)**



83. **Ans (4)**

The middle figure inverts and one part becomes shaded.

So, in answer image no shaded part and inverted middle part must be there.

Reflection of Z as seen in water is taken and written on right side in 2nd figure.

Same way reflection of 6 as seen in water is taken and written on left side in 2nd figure.

Use the inverse of this to get the answer as option d i.e. answer figure 4.

84. **Ans (2)**

Consider one column at a time. Fishes are moving 90° clockwise.

Also, their tail inverts every time they move by 90°.

Move the fish by 90° and then invert the tail.

So, answer is option 2.

85. **Ans (1)**

True progress in technology involves some risks too

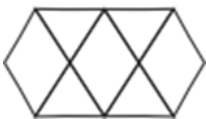
86. **Ans (4)**

Measures have to be taken to wisely use the available natural resources

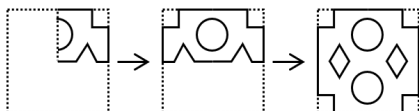
87. **Ans (1)**

Only in the first series, the image rotates 45 degrees as it moves towards the right side. In the second series, it turns 30 degrees; in third series, it rotates 90 degrees, and in the fourth series, it rotates by 15 degrees as it moves one step forward towards the right side. See the corner of the image, which forms a small square, to understand the movement of the image.

88. **Ans (2)**



89. Ans (3)



90. Ans (2)

The average of two numbers is written above them.

$$(20 + 24) / 2 = 22$$

$$(24 + 32) / 2 = 28$$

$$(32 + 36) / 2 = 34$$

$$(22 + 28) / 2 = 25$$

$$(28 + 34) / 2 = 31$$

$$(25 + 31) / 2 = 28$$

Hence, option 2 is the correct answer.

91. Ans (1)

Clearly t can be any real number.

$$\text{Let } t = \tan \theta \Rightarrow x = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \Rightarrow x = \cos 2\theta \text{ and } y = \frac{2 \tan \theta}{1 + \tan^2 \theta} = \sin 2\theta \Rightarrow x^2 + y^2 = 1$$

$$\text{Thus required area} = \pi \cdot l^2 = \pi \text{ sq. units}$$

92. Ans (3)

$$f'(a) = 2, f(a) = 4$$

$$\lim_{x \rightarrow a} \frac{xf(a) - af(x)}{x - a}$$

$$\Rightarrow \lim_{x \rightarrow a} \frac{f(a) - af'(x)}{1} \quad (\text{Lopitals rule})$$

$$= f(a) - af'(a)$$

$$= 4 - 2a$$

93. Ans (4)

$$\text{Put } 2x = \tan \theta$$

$$I = \frac{1}{2} \int_0^{\pi/4} \ln(1 + \tan \theta) d\theta$$

$$\therefore I = \frac{\pi}{16} \ln 2$$

94. Ans (3)

$$\lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n}\right)^a + \left(1 + \frac{2}{n}\right)^a + \dots + \left(1 + \frac{n}{n}\right)^a}{\left(\frac{1}{n}\right)^a + \left(\frac{2}{n}\right)^a + \dots + \left(\frac{n}{n}\right)^a}$$

$$= \lim_{n \rightarrow \infty} \frac{\int_0^1 (1+x)^a dx}{\int_0^1 x^a dx} = \frac{\frac{(1+x)^{a+1}}{a+1} \Big|_0^1}{\frac{x^{a+1}}{a+1} \Big|_0^1}$$

$$= 2^{a+1} - 1 = 15$$

$$2^{a+1} = 16$$

$$a = 3$$

95. Ans (2)

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

$$\Rightarrow (a+b+c)(ab+bc+ca-a^2-b^2-c^2)=0$$

Since $a+b+c \neq 0$ for any triangle.

$$\text{We } ab+bc+ca=a^2+b^2+c^2$$

$$\Rightarrow a=b=c$$

$$A=B=C=60^\circ$$

$$\Rightarrow \sin A \sin B + \sin B \sin C + \sin C \sin A = \sin^2 A + \sin^2 B + \sin^2 C$$

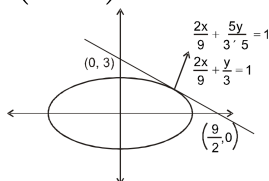
$$= \frac{3}{4} + \frac{3}{4} + \frac{3}{4} = \frac{9}{4}$$

96. Ans (4)

$$\frac{x^2}{9} + \frac{y^2}{5} = 1$$

$$a=3, b=\sqrt{5}$$

$$\left(ae, \frac{b^2}{a} \right)$$



$$\frac{b^2}{a} = \frac{5}{3}$$

$$e = \sqrt{1 - \frac{5}{9}} = \frac{2}{3}$$

$$\left(ae, \frac{b^2}{a} \right) = \left(2, \frac{5}{3} \right)$$

$$\text{Area} = 4 \left(\frac{1}{2} \times \frac{9}{2} \times 3 \right) = 27$$

97. Ans (3)

$$f(x) = \cos[\pi^2]x + \cos[-\pi^2]x$$

we know $9 < \pi^2 < 10$ and $-10 < -\pi^2 < -9$

$$\Rightarrow [\pi^2] = 9 \text{ and } [-\pi^2] = -10$$

$$\therefore f(x) = \cos 9x + \cos(-10x)$$

$$f(x) = \cos 9x + \cos 10x$$

$$(a) f\left(\frac{\pi}{2}\right) = \cos \frac{9\pi}{2} + \cos 5\pi = -1 \text{ (false)}$$

$$(b) f(\pi) = \cos 9\pi + \cos 10\pi = -1 + 1 = 0 \text{ (false)}$$

$$(c) f(-\pi) = \cos(-9\pi) + \cos(-10\pi)$$

$$= \cos 9\pi + \cos 10\pi = -1 + 1 = 0 \text{ (true)}$$

$$(d) f\left(\frac{\pi}{4}\right) = \cos \frac{9\pi}{4} + \cos \frac{5\pi}{2} = \cos\left(2\pi + \frac{\pi}{4}\right) + 0$$

$$= \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} \text{ (false)}$$

98. **Ans (2)**

$$f(x) = |\sin 4x| + |\cos 2x|$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ \text{Period} = \frac{\pi}{4} & & \text{Period} = \frac{\pi}{2} \end{array}$$

$$\text{Fundamental period} = \frac{\pi}{2}$$

99. **Ans (4)**

e_1 and e_2 are the eccentricities of the ellipse, $x^2/18 + y^2/4 = 1$ and the hyperbola, $x^2/9 - y^2/4 = 1$ respectively

$$e_1 = \sqrt{1 - \frac{4}{18}} = \frac{\sqrt{7}}{3}$$

$$e_2 = \sqrt{1 + \frac{4}{9}} = \frac{\sqrt{13}}{3}$$

As (e_1, e_2) lies on the ellipse $15x^2 + 3y^2 = k$

$$\text{Because, } 15e_1^2 + 3e_2^2 = k$$

$$\Rightarrow 15 \times (7/9) + 3 \times (13/9) = k$$

$$\Rightarrow k = 16$$

100. **Ans (4)**

$$x = 5 \tan \phi, y = 4 \sec \phi$$

$$\Rightarrow \frac{y^2}{16} - \frac{x^2}{25} = 1$$

$$e = \sqrt{1 + \frac{a^2}{b^2}} = \sqrt{1 + \frac{25}{16}}$$

101. **Ans (2)**

$$= \tan[\tan^{-1} a_2 - \tan^{-1} a_1 + \tan^{-1} a_3 - \tan^{-1} a_2 + \dots + \tan^{-1} a_n - \tan^{-1} a_{n-1}]$$

$$= \tan[\tan^{-1} a_n - \tan^{-1} a_1]$$

$$= \tan \left[\tan^{-1} \frac{a_n - a_1}{1 + a_1 a_n} \right]$$

$$\therefore \frac{a_n - a_1}{1 + a_1 a_n} = \frac{(n-1)d}{1 + a_1 \cdot a_n}$$

102. **Ans (3)**

$$\lim_{x \rightarrow \alpha} \frac{1 - \cos(a(x-\alpha)(x-\beta))}{(x-\alpha)^2}$$

$$= \lim_{x \rightarrow \alpha} \left(\frac{1 - \cos(a(x-\alpha)(x-\beta))}{(a(x-\alpha)(x-\beta))^2} \right) \frac{(a(x-\alpha)(x-\beta))^2}{(x-\alpha)^2}$$

$$= \frac{1}{2}(a^2)(\alpha-\beta)^2 \left(\text{using } \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2} = \frac{1}{2} \right)$$

103. Ans (2)

$$\text{Since, } \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2} \right)^{2x} = e^2$$

$$\therefore \lim_{x \rightarrow \infty} \left[\left(1 + \frac{ax+b}{x^2} \right)^{\frac{x^2}{ax+b}} \right]^{\frac{2(ax+b)}{x}} = e^2$$

$$\Rightarrow \lim_{x \rightarrow \infty} e^{\frac{2(ax+b)}{x}} = e^2$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{2(ax+b)}{x} = 2$$

$$\Rightarrow 2a = 2$$

$$\Rightarrow a = 1$$

Thus $a = 1$ and $b \in \mathbb{R}$

104. Ans (4)

$$\Rightarrow |A|^2 = \frac{1}{25}$$

$$|5\text{adj}(A)| = 5 \Rightarrow 5^3 |A|^2 = 5 \Rightarrow |A| = \pm \frac{1}{5}$$

105. Ans (4)

$$A^2 = 3A + \alpha I$$

$$A^3 = 3A^2 + \alpha A$$

$$A^3 = 3(3A + \alpha I) + \alpha A$$

$$A^3 = 9A + \alpha A + 3\alpha I$$

$$A^4 = (9 + \alpha)A^2 + 3\alpha A$$

$$= (9 + \alpha)(3A + \alpha I) + 3\alpha A$$

$$= A(27 + 6\alpha) + \alpha(9 + \alpha)$$

$$\Rightarrow 27 + 6\alpha = 21 \Rightarrow \alpha = -1$$

$$\Rightarrow \beta = \alpha(9 + \alpha) = -8$$

106. Ans (4)

Required probability

$$\Rightarrow P = \frac{\frac{1}{3} \times \frac{1}{5}}{\frac{1}{3} \times \frac{1}{5} + \frac{1}{3} \times \frac{1}{6} + \frac{1}{3} + \frac{1}{7}} = \frac{42}{107}$$

107. Ans (1)

Let E_1 = Event that a student fail in Physics

E_2 = Event that a student fail in Mathematics

$$\text{Then, } P(E_1) = \frac{30}{100} = \frac{3}{10}, P(E_2) = \frac{25}{100} = \frac{1}{4}$$

$$P(E_1 \cap E_2) = \frac{10}{100} = \frac{1}{10}$$

Hence, the required probability

$$= P(E_1/E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)}$$

$$= \frac{1/10}{1/4} = \frac{4}{10} = \frac{2}{5}$$

108. **Ans (3)**

Fixing two 'S' s one in the beginning and the other in the end, we are left with eight letters of which T is repeated thrice, I is repeated twice. Hence, the required number of ways

$$= \frac{8!}{3!2!} = \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2}{12} = 3360$$

109. **Ans (4)**

Case – I, digits used are 1,2,3,4,5, $5! = 120$

Case – II, when digits used are 0,1,4,5,2 $= 5! - 4! = 96$

$$\therefore \text{Total} = 120 + 96 = 216$$

110. **Ans (3)**

eqⁿ of parabola is

$$(y-4)^2 = -8(x-5)$$

$$y^2 + 16 - 8y = -8x + 40$$

$$y^2 + 8x - 8y - 24 = 0$$

111. **Ans (4)**

Given parabola can be written as $(y+1)^2 = -(x-1)$

112. **Ans (1)**

$$\begin{array}{c} \text{1 : 2} \\ \text{O} \quad \text{G} \quad \text{H} \\ \text{(h,k)} \quad \text{(3,3)} \quad \text{(-3,5)} \\ \text{G divides OH in ratio 1 : 2} \\ 3 = \frac{-3+2h}{3}, 3 = \frac{5+2k}{3} \\ \therefore h = 6, k = 2 \end{array}$$

113. **Ans (4)**

$p + q = \text{sum of the roots} = -p$

$$\Rightarrow 2p + q = 0 \quad \dots(1)$$

and $pq = \text{product of the roots} = q$

$$\Rightarrow q(p-1) = 0 \Rightarrow p = 1 \quad (\because q \neq 0)$$

Then, from (1), $2 + q = 0 \Rightarrow q = -2$.

114. **Ans (3)**

Required line is

$$(2x + y - 1) + \lambda(3x + 2y - 5) = 0$$

it passes through (0, 0), then

$$-1 + \lambda(-5) = 0$$

$$\Rightarrow \lambda = -\frac{1}{5}$$

115. Ans (3)

Given $\theta = 135^\circ$ & y intercept = -5

then $m = \tan 135^\circ = -1$ & $c = -5$

$\therefore$ equation of line $y = -1 \cdot x - 5$

$$\Rightarrow x + y + 5 = 0.$$

116. Ans (3)

Given

$$n(\text{coffee}) = 37$$

$$n(\text{tea}) = 52$$

$$n(\text{tea} \cup \text{coffee}) = 70$$

$$n(\text{tea} \cup \text{coffee}) = n(\text{coffee}) + n(\text{tea}) - n(\text{tea} \cap \text{coffee})$$

$$70 = 37 + 52 - n(\text{tea} \cap \text{coffee})$$

$$n(\text{tea} \cap \text{coffee}) = 37 + 52 - 70$$

$$= 89 - 70$$

$$= 19$$

117. Ans (4)

$$n(A \cup B \cup C) = 3 + 3 + 3 - 2 - 2 - 2 + 1 = 4$$

118. Ans (3)

$$a + 2d = 7 \quad \dots(1)$$

$$T_7 = 2 + 3T_3$$

$$\Rightarrow a + 6d = 2 + 3(7) = 23 \quad \dots(2)$$

from (1) and (2)

$$d = 4 \text{ and } a = -1$$

$$\therefore S_{20} = \frac{20}{2} [2(-1) + 19(4)] = 740$$

119. Ans (2)

$$\sum_{i=1}^{10} (x_i - 50)^2 = 250$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{10} (x_i - 50)^2}{10}} = 5$$

$$\text{Coeff. of variation} = \frac{\sigma}{x} \times 100 = 10\%$$

120. Ans (2)

Here $a = \sqrt{4+4+1} = \sqrt{9}$, $b = \sqrt{1+4+4} = \sqrt{9}$ and $c = \sqrt{1+16+1} = \sqrt{18}$.

Obviously, it is a right angled and isosceles triangle.

121. Ans (3)

$$\sqrt{2} \sec \theta + \tan \theta = 1$$

$$\sqrt{2} \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} = 1$$

$$\Rightarrow \sqrt{2} + \sin \theta = \cos \theta$$

$$\cos \theta - \sin \theta = \sqrt{2}$$

$$\Rightarrow \frac{1}{\sqrt{2}} \cos \theta - \frac{1}{\sqrt{2}} \sin \theta = 1$$

$$\cos \left(\theta + \frac{\pi}{4} \right) = \cos 0 \Rightarrow \theta + \frac{\pi}{4} = 2n\pi \pm 0$$

$$\theta = 2n\pi - \frac{\pi}{4}$$

122. Ans (3)

$$n(m^2 - 1) = (\sec \theta + \operatorname{cosec} \theta) \cdot 2 \sin \theta \cos \theta$$

$$\therefore m^2 = 1 + 2 \sin \theta \cos \theta$$

$$= \frac{\sin \theta + \cos \theta}{\sin \theta \cdot \cos \theta} \cdot 2 \sin \theta \cos \theta = 2m$$

123. Ans (4)

$$AB = (\hat{i} - 3\hat{j}) - (2\hat{i} + \hat{j}) = -\hat{i} - 4\hat{j};$$

$$CD = (\hat{i} + \lambda\hat{j}) - (3\hat{i} + 2\hat{j}) = -2\hat{i} + (\lambda - 2)\hat{j};$$

$$AB \parallel CD \Rightarrow AB = x \cdot CD$$

$$-\hat{i} - 4\hat{j} = x \{ -2\hat{i} + (\lambda - 2)\hat{j} \}$$

$$\Rightarrow -1 = -2x, -4 = (\lambda - 2)x$$

$$\Rightarrow x = \frac{1}{2} \text{ and } \lambda = -6$$

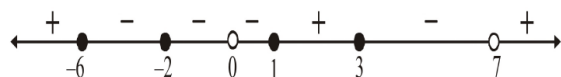
124. Ans (3)

$$\frac{\sin \alpha}{\sin \beta} = \frac{5}{3}$$

$$\frac{\sin \alpha + \sin \beta}{\sin \alpha - \sin \beta} = \frac{5 + 3}{5 - 3} = 4$$

125. Ans (2)

We mark on the number line zeroes of numerator of expression : 1, -2, 3 and -6 (with black circles) and the zeroes of denominator 0 and 7 (with white circles), isolate the double points : -2 and 0 and draw the wavy curve :



$$x \in (-\infty, -6] \cup \{-2\} \cup [1, 3] \cup (7, \infty)$$

126. Ans (3)

$$\left(2x + \frac{1}{3x}\right)^6$$

$$\text{General term } T_{r+1} = {}^6C_r (2x)^{6-r} \left(\frac{1}{3x}\right)^r$$

term independent of x

$$6 - r - r = 0$$

$$2r = 6$$

$$r = 3$$

$$T_4 = {}^6C_3 2^3 \cdot \frac{1}{3^3} = \frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} \cdot \frac{2 \cdot 2 \cdot 2}{3 \cdot 3 \cdot 3} = \frac{160}{27}$$

127. Ans (4)

$$f(x) = \begin{cases} 2x + \tan^{-1} x + b & x < 0 \\ x^3 + x^2 + ax + c & x \geq 0 \end{cases}$$

$$\because \text{Continuous at } x = 0 \Rightarrow b = c \Rightarrow \frac{b}{c} = 1$$

$$\because \text{differentiable at } x = 0$$

$$\Rightarrow 2 + 1 = a \Rightarrow a = 3$$

$$\frac{b^2}{c^2} + a = 4$$

128. Ans (2)

$$\text{Putting } z = x + iy \text{ and } \arg(z_1/z_2) = \arg(z_1) - \arg(z_2),$$

we get

$$\tan^{-1} \frac{y}{x-1} - \tan^{-1} \frac{y}{x+1} = \frac{\pi}{4}$$

$$\text{or } \frac{[y/(x-1)] - [y/(x+1)]}{1 + [y^2/(x^2-1)]} = \tan \frac{\pi}{4} = 1$$

$$\text{or } 2y = x^2 + y^2 - 1$$

$$\text{or } x^2 + y^2 - 2y - 1 = 0 \Rightarrow \text{circle}$$

129. Ans (2)

$$\Rightarrow 10 + 3.4 + x = 9m \Rightarrow x = 5$$

130. Ans (3)

$$(p \wedge q) \vee \sim p = (p \vee \sim p) \wedge (q \vee \sim p)$$

$$= t \wedge (q \vee \sim p) = q \vee \sim p = \sim p \vee q$$